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Interlanguage Articles: Bound or Free?

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1. Introduction*

A significant body of research in the field of second language (L2) acquisition had focused on the intractable difficulties that L2 learners have with functional morphology. Although most research in this area has provided syntactic, semantic or pragmatic solutions to these problems (e.g. Ionin, Ko & Wexler 2004, Trenkic 2007, Tsimpli 2003), we have taken a different tack. We have proposed the Prosodic Transfer Hypothesis (PTH) which argues that the difficulties that learners have with the production of function words and inflectional morphology stem from constraints on prosodic structure that are transferred from the native (L1) grammar: functional material may sometimes be omitted or pronounced in a non-target manner if the necessary prosodic representations are not available in the L1 (Goad, White & Steele 2003, Goad & White 2004, 2006).

In the domain of article acquisition, previous research relating to the PTH has shown that L1 prosodic constraints result in deletion of articles, production of articles as stressed or production of articles as seemingly target-like under certain conditions (Goad & White 2008, 2009a,b). Aside from appeals to proficiency level, however, no attempt has been made to predict when each type of output will occur. In this paper, we argue that learners' outputs reflect L1-based assumptions about the lexical status of the article, as [±bound]. This, in turn, affects how the article is prosodified. Our focus is on Turkish-speaking learners of English.

2. Representations

We begin by motivating the prosodic representations for articles and other determiners in English and Turkish. The presence or absence of word order alternations in DPs with and without adjectives will be of central importance to the proposed cross-linguistic differences in prosodification. These differences in word order and prosodification will be argued to stem from the [±bound] status of articles in the two languages.

2.1. Prosodic representation of determiners in English and Turkish

We assume that prosodic constituents are organized into the hierarchy in (1) (e.g. Selkirk 1980, 1986, McCarthy & Prince 1986, Nespor & Vogel 1986). Normally, lexical words form their own prosodic words while unstressed functional material is organized so as to be dependent on the prosodic structure of its host.

(1) Phonological Phrase (PPh)
|
Prosodic Word (PWd)
|
Foot (Ft)
|
Syllable (σ)

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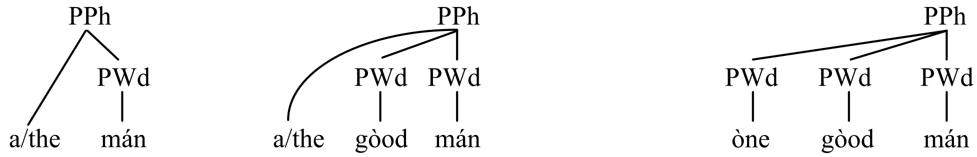
Articles in English are organized as free clitics, linking directly to the PPh that dominates their host (Selkirk 1996); see (2a). This representation permits adjectives to intervene between article and noun, as can be seen in (2b). Other determiners are normally stressed and thus form independent PWds; see (2c).

(2) English:

a. Articles: b. Articles in DPs with adjectives: c. Other determiners:

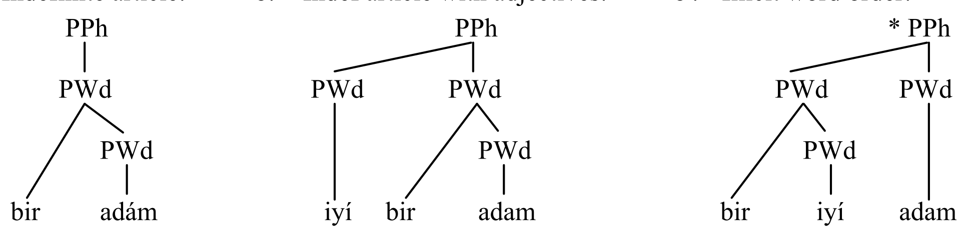


Turkish has one article, indefinite *bir*. Although *bir* is unstressed, as are English articles, it is not organized as a free clitic. Rather, it is an affixal clitic (prefix), adjoined to the PWd of its host noun; see (3a) (Goad & White 2004). Evidence for this structure over the alternative (observed in English) is that, in DPs with adjectives, the adjective must appear to the left of the article to enable the article to prefix onto the noun, as shown in (3b).[1] The illicit structure in (3b'), where *bir* prefixes onto the adjective, may be permitted in some languages (e.g. Bulgarian; see Embick & Noyer 2001 for discussion consistent with this) but it is illicit in Turkish because APs are late adjoined (see section 2.2); in other words, the adjective is not available early enough in the derivation to enable *bir* to prefix onto it.[2]

When stressed, *bír* is interpreted as the numeral 'one' and, by virtue of its stress, it forms an independent PWd, as seen in (3c). Since stressed determiners form their own PWds, they are not prosodically dependent on the following noun; hence adjectives can intervene. In other words, the canonical word order for Turkish, Det-Adj-N, is observed in this construction.

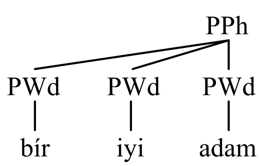
(3) Turkish:[3]

a. Indefinite article: b. Indef article with adjectives: b'. Illicit word order:



'a man' 'good a man' 'a good man'

c. Other determiners:



'one good man'

[1] See Poser (1990) for a similar analysis of Aoyagi pre-nominal modifiers in Japanese, some of which have the semantics of determiners.

[2] Thanks to Dave Embick for helpful discussion of (3b').

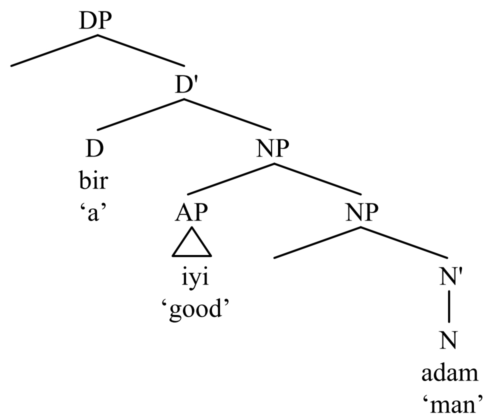
[3] The Turkish examples containing more than one PWd only have phrase-level stress marked because the cues to word-level stress on other PWds in the PPh are weak. We have followed Kabak & Vogel (2001) who state that phrase-level stress falls on the final syllable of the leftmost PWd in the phrase, as this corresponds to the judgments received from the native speakers we consulted (see Goad & White 2009a). See Inkelas & Orgun (2003) for a different view.

2.2. Word order alternations in Turkish

As mentioned above, the canonical word order for Turkish DPs is Det-Adj-N and, thus, some type of movement is required to yield the non-canonical order observed for DPs with indefinite articles, shown in (3b). We assume that post-syntactic local dislocation is involved (Embick & Noyer 2001), motivated by the [+bound] status of the Turkish article (see below).

We first sketch a syntactic analysis of the Turkish DP. Following Goad and White (2009b), we assume that APs are adjuncts in Turkish, as in (4).

(4)



'a good man'

This analysis is supported by the fact that ordering of adjectives in DPs is relatively free (i.e., in DPs not containing stressed or unstressed *bir* (Öner Özçelik, personal communication)); see the examples in (5) from Öztürk (2005).

(5) a.	bu kırmızı kitap this red book	kırmızı bu kitap red this book	'this red book'
b.	iki kırmızı kitap two red book	kırmızı iki kitap red two book	'two red books'

If APs are late adjoined (after spell out of the DP), the determiner and noun will be string adjacent at PF and local dislocation can take place. Local dislocation allows [bir+N] to form a morphological word which, in turn, leads to the prosodification of indefinite *bir* as an affixal clitic.

We propose that the [±bound] status of determiners is responsible for the presence or absence of word order alternations in DPs containing adjectives, as well as for the associated prosodifications in (2) and (3). As mentioned above, we assume that local dislocation of indefinite *bir* is motivated by its being marked as [+bound] in the Vocabulary, as indicated in (6a). This status requires that the article form a morphological word with the following noun, through PF movement as discussed. English articles, by contrast, are [-bound] (see (6b)) and, thus, require no post-syntactic movement. This is why there is no difference in word order in English DPs with and without adjectives. The [-bound] status of English articles, combined with lack of stress, results in their prosodification as free clitics, shown earlier in (2a-b). Other determiners in both languages are also [-bound], as indicated in (6c-d); this, combined with their status as stressed, results in their prosodification as independent PWds, as seen in (2c) and (3c).

(6) [±bound] status of determiners:

Articles:	a.	Turkish:	[+bound]
	b.	English:	[-bound]
Other determiners:	c.	Turkish:	[-bound]
	d.	English:	[-bound]

3. Experiment

For Turkish-speaking learners of English, target production of articles in DPs with and without adjectives requires acquisition of two properties: the [-bound] lexical entry in (6b) and the free clitic structure in (2a-b). Below, we report on an experiment that examines the interaction of these properties in the interlanguage grammar.

3.1. Methodology

The data reported on here were collected as part of a larger study which examined the acquisition of DPs by 18 Turkish-speaking learners of English (see Goad & White 2009a,b). Their proficiency levels, determined by means of a cloze test and self-report, were as follows: low (n=9), intermediate (n=7), advanced (n=2). Six subjects were tested in Montreal and 12 in Turkey.

Subjects were taped on a SONY PCM-M1 DAT recorder describing a sequence of pictures, designed to elicit DPs with and without adjectives. The data were coded for a number of syntactic measures, including presence or absence of articles, substitutions, which article would be expected in the context, etc. The data were also narrowly transcribed with particular attention paid to vowel quality and the phonetic correlates of stress.

In this paper, we report on the 15 subjects who produced at least seven definite or seven indefinite DPs with adjectives.[4,5]

3.2. Predictions

As mentioned above, the target representation for English articles requires acquisition of the [-bound] lexical entry and free clitic structure in (2a-b). Predicted interlanguage patterns are provided in (7).

(7) Predicted interlanguage patterns:

Patterns:		Structure for article:	DPs without adj:	DPs with adj:
A	Transfer representation for indefinite *bir*	[+bound] affixal clitic	unstressed article	deletion of article
B	Transfer representation for other determiners	[-bound] independent PWd	stressed article	stressed article
C	Acquire target representation	[-bound] free clitic	unstressed article	unstressed article

Concerning Pattern A, if L2 speakers transfer the [+bound] analysis of indefinite *bir* (see (3a-b)) into the interlanguage English grammar, they should produce target-*sounding* articles in DPs without adjectives, even if the target analysis and corresponding prosodification have not in fact been acquired. In adjective contexts, though, article deletion is predicted. Articles cannot prefix onto adjectives in Turkish, as would be required if this analysis were applied to English word order, exemplified by the following: *[a $[\text{good}]_{\text{PWd}}]_{\text{PWd}}$ $[\text{man}]_{\text{PWd}}]_{\text{PPh}}$. As discussed, the [+bound] analysis and corresponding prosodification is only compatible with Adj-Art-N which is illicit in English.

Turning to Pattern B, if L2 speakers transfer the [-bound] status of other determiners into the interlanguage (see (3c)), stressing of articles is predicted, regardless of whether or not the DP contains adjectives. Stressing is predicted because other determiners form independent PWds in Turkish (just as they do in English).

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[4] One subject, T4, produced only six indefinite DPs with adjectives. We have nonetheless included this subject because she treated all six DPs in the same fashion: the article was deleted.

[5] The proficiency levels of these 15 subjects are as follows: low: T4, T7, T10, T11, T14, T17; intermediate: T1, T3, T8, T9, T13, T15, T16; high: T2, T6.

Finally, if L2 learners have successfully acquired the [-bound] representation and corresponding prosodification in (2a-b) for English, unstressed articles are predicted, regardless of whether or not the DP contains adjectives.

3.3. Results and discussion

The results are provided in Tables 1 and 2 below. The patterns found for definite and indefinite DPs are presented separately, as most subjects do not treat the two articles uniformly; we return to this below. The boxed numbers in the first column of both tables reveal that some subjects (T10, T11 in Table 1; T4, T16 in Table 2) show evidence of a Stage Ø, not included in the predicted patterns in (7), where the predominant pattern is deletion of articles in both types of DPs. We will not discuss across-the-board deletion and will instead focus on the other patterns of behaviour exhibited.

For definite DPs, Table 1 shows that two patterns are observed. Seven subjects (T3, T6, T8, T13, T15, T16, T17) exhibit behaviour consistent with patterns A and C. We interpret this to indicate that these subjects are in transition from one stage (A) where they have transferred the representation for indefinite *bir* into the interlanguage grammar, thereby yielding target-*sounding* (unstressed) articles in DPs without adjectives but article deletion in DPs with adjectives, to another stage (C), the target grammar, where they are correctly producing unstressed articles in both types of DPs. These subjects are thus wavering between a [+bound] and [-bound] representation for English articles. Four other subjects (T1, T2, T9, T14) have arrived at the target grammar, producing a considerable number of articles in target fashion. The behaviour of the remaining subject in Table 1 (T7) does not fit with our predictions.

		Ø		A→C		C		?	
		T10 T11		T3 T6 T8 T13 T15 T16 T17		T1 T2 T9 T14		T7	
		#	%	#	%	#	%	#	%
No adj	unstressed	11	24.5	450	70.6	189	76.8	20	50.0
	stressed	6	13.3	33	9.3	23	9.4	10	25.0
	deleted	28	62.2	71	20.1	34	13.8	10	25.5
Adj	unstressed	1	5.3	77	50.7	60	83.3	2	22.2
	stressed	0	0	26	17.1	9	12.5	5	55.6
	deleted	18	94.7	49	32.3	3	4.2	2	22.2

Table 1. Patterns of article production: Definite DPs.

Turning to indefinite DPs, Table 2 similarly reveals two patterns of behaviour. Interestingly, however, the non-target behaviour exhibited for indefinites is different from that exhibited for definites. Six subjects (T3, T6, T8, T9, T10, T14) follow pattern B: a majority of indefinite articles are produced as stressed in DPs both with and without adjectives. Two subjects (T1, T2) produce unstressed articles in both contexts and have, thus, acquired the target grammar.

All eight of these subjects have acquired the [-bound] representation for English articles. However, they differ in how they prosodify the article. There are two possible prosodifications that stem from a [-bound] representation for a function word, PWd or free clitic. Only one is sanctioned by the Turkish grammar: [-bound] determiners, i.e. determiners other than articles, are stressed and thereby prosodified as independent PWds. Both prosodifications are sanctioned by the English grammar; however, the prosodification for [-bound] articles (as opposed to other determiners) is the free clitic representation. This representation has been acquired by the two subjects exhibiting pattern C but clearly not by those who follow pattern B.

		Ø		B		C	
		T4 T16		T3 T6 T8 T9 T10 T14		T1 T2	
		#	%	#	%	#	%
No adj	unstressed	5	10.4	41	19.2	64	79.0
	stressed	16	33.3	131	61.2	10	12.3
	deleted	27	56.3	42	19.6	7	8.7
Adj	unstressed	0	0	13	14.6	33	73.3
	stressed	2	11.8	57	64.0	5	11.1
	deleted	15	88.2	19	21.4	7	15.6

Table 2. Patterns of article production: Indefinite DPs.

The predictions laid out in (7) are largely confirmed: subjects who produce unstressed articles in (definite) DPs without adjectives either delete articles in DPs with adjectives (A) or produce them in target fashion (C); subjects who stress articles in (indefinite) DPs without adjectives do so in DPs with adjectives as well (B); and subjects who have acquired the target representation produce unstressed articles in DPs both with and without adjectives (C). Only one subject, T7, behaves somewhat anomalously.

There is, however, a robust definite-indefinite asymmetry which was not predicted. A is the predominant non-target pattern for definites: *the* is a [+bound] affixal clitic, the same representation as holds for indefinite *bir*. B is the predominant non-target pattern for indefinites: *a* is a [-bound] independent PWd, like other determiners in Turkish.

We reason that the explanation for this asymmetry stems from learners' assumptions about the underlying vowel quality for each article. Our earlier exploration of vowel harmony targeting articles in the grammars of a different subset of the same 18 subjects revealed that the default vowel in the definite article is typically central ([ɨ] or [ə]), while the default vowel in the indefinite article is front ([ɪ], [e] or [ɛ]) (Goad & White 2009b). This difference likely stems from English orthography: orthographic *a* is often a front vowel, [ei], notably in the letter *A*, while orthographic *e* is often a schwa-like vowel ([ə] or [ɨ]). If *the* contains a schwa-like vowel, a vowel which is unstressable in most languages including English,[6] then the only suitable representation available for transfer from the L1 grammar is the representation for unstressed indefinite *bir*. That is, with an underlying schwa-like vowel, the representation for other (stressed) determiners in Turkish is simply not appropriate for *the*. By contrast, since orthographic *a* is often realized as [ei] in English and this vowel is typically stressed, speakers may have concluded that the [-bound] independent PWd representation for other determiners in Turkish is most appropriate for the indefinite article in English.

4. Conclusion

In conclusion, consistent with our previous work (e.g. Goad and White 2008, 2009a, b), we have suggested that Turkish-speaking L2ers have recourse to different ways of representing the prosodic structure of English articles in the interlanguage grammar, based on what is permissible in the L1. Depending on the prosodic representation adopted, articles will be deleted, produced as seemingly target-like or produced as stressed. In this paper, we have extended our analysis to look not just at the prosodic representations employed but also at the interlanguage analysis of articles as bound or free. In consequence, a clear set of predictions emerges as to the conditions under which particular non-target outputs should occur. For learners who analyse articles as [+bound], and thereby transfer the affixal clitic representation into the interlanguage grammar, DPs without adjectives should contain articles that sound native-like (unstressed) but DPs with adjectives should exhibit article deletion. For learners who analyse articles as [-bound], and thereby transfer the independent PWd representation for other

[6] Central vowels are absent from Turkish.

determiners into the interlanguage grammar, non-target productions should be limited to articles that are stressed, regardless of the shape of the DP. These predictions are largely confirmed.

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